Research Scientist, Biomedical Optics, Levi Lab

UT Southwestern Medical Center · Dallas, TX, US

Job ID (this archive)	HI-034
Employer	UT Southwestern Medical Center
Location(s)	Dallas, TX, US
Career stage (archive tag)	Research / graduate degree
Location group	DFW area
Majors (archive tags)	Biomedical Engineering, Mechanical Engineering, Data Science
Pay listed	Not listed
Employment type	Full-time
Employer requisition	968579
Recruiting status when checked	Open when checked (Oct 7, 2026). Employer note: Readable; no closure notice found Saved PDF = full employer posting text, reformatted for print.
Source page	https://jobs.utsouthwestern.edu/job/23767000/research-scientist-biomedical-optics-levi-lab-dallas-tx/
Captured	2026-10-07T20:11:31.745316 UTC
Course sheet refs	HI-034

> Captured on 2026-10-07 from the employer's public career site. The posting text below is the employer's own wording, reformatted for printing (site navigation removed). Live listings change and close; check the source page before applying.

Job description

Research Scientist, Biomedical Optics, SWIR Imaging, and Burn-Wound Assessment, Levi Lab

WHY UT SOUTHWESTERN?

With over 75 years of excellence in Dallas-Fort Worth, Texas, UT Southwestern is committed to excellence, innovation, teamwork, and compassion. As a world-renowned medical and research center, we strive to provide the best possible care, resources, and benefits for our valued employees. Ranked as the number 1 hospital in Dallas-Fort Worth according to U.S. News & World Report , we invest in you with opportunities for career growth and development to align with your future goals. Our highly competitive benefits package offers healthcare, PTO and paid holidays, on-site childcare, wage, merit increases and so much more. We invite you to be a part of the UT Southwestern team where you'll discover a culture of teamwork, professionalism, and a rewarding career!

JOB SUMMARY

Plans, coordinates, directs, and participates in work of smaller research project or major unit of large research project in field of specialization for which incumbent is particularly qualified in terms of extensive training and/or experience.

The Levi Laboratory seeks a Research Scientist to lead the development and translational application of optical imaging technologies for objective assessment of skin viability, burn-wound depth, tissue composition, and healing. The laboratory integrates visible, near-infrared (NIR), and short-wave infrared (SWIR) imaging with porcine injury models, human clinical studies, quantitative histology, biostatistics, and machine learning.

The successful candidate will design, build, calibrate, operate, and refine multispectral imaging systems; acquire high-quality images in laboratory, animal, operating-room, and clinical environments; develop reproducible image-processing and classification pipelines; and correlate optical measurements with histopathology and clinical assessments. The position requires the ability to bridge optical engineering, tissue optics, image analysis, experimental biology, and clinical translation. A commitment of at least two years is preferred.

RESEARCH PROGRAM

Multispectral SWIR reflectance imaging of burn wounds using the laboratory's SWIR assessment platform and bands centered near 1200, 1650, 1940, and 2250 nm for sensitivity to water, lipids, collagen, and tissue scattering.

Objective differentiation of superficial, deep partial-thickness, and full-thickness burns, including identification of wounds likely to require operative debridement or grafting.

Translation of a SWIR assessment platform from tissue phantoms and porcine burn models to imaging of human patients in the operating room.

Co-registration of visible, multispectral optical, histologic, and clinical data to establish reliable biological and clinical ground truth.

Development of reflectance indices, quantitative optical-property maps, statistical classifiers, and machine-learning models for tissue viability and burn-depth prediction.

Expansion toward hyperspectral imaging, spatial frequency domain imaging, tissue spectroscopy, fluorescence imaging, and other emerging SWIR biomedical technologies when scientifically appropriate.

ESSENTIAL JOB DUTIES

Design, assemble, align, and maintain visible, NIR, multispectral SWIR, or hyperspectral imaging systems using broadband or wavelength-specific sources, diffusers, collimators, dichroic optics, bandpass filters, filter wheels, lenses, apertures, spectrometers, and silicon, germanium, or InGaAs detectors.

Select wavelengths, optical components, acquisition geometry, exposure or integration times, gain, working distance, field of view, spatial resolution, and illumination parameters based on tissue optical properties and the intended biological measurement.

Calibrate imaging systems using reflectance standards, dark and white references, tissue-simulating phantoms, and wavelength-appropriate correction procedures; characterize illumination uniformity, detector response, noise, repeatability, drift, and signal-to-noise ratio.

Develop and execute standardized imaging protocols for tissue phantoms, ex vivo samples, in vivo porcine burn models, and human burn patients, including operating-room workflows that maintain safety, sterility, patient privacy, and minimal disruption to clinical care.

Acquire synchronized visible and SWIR images or videos and accurately document image settings, subject positioning, anatomical location, burn timing, treatment status, calibration data, and relevant environmental variables.

Perform image preprocessing, including bad-pixel and dark-current correction, flat-field or reference normalization, frame averaging, denoising, motion correction, distortion correction, image registration, segmentation, and quality-control review.

Define and track anatomically matched regions of interest across visible images, SWIR wavelength bands, serial time points, surgical markings, biopsies, and histologic sections.

Quantify wavelength-specific and normalized reflectance, spectral ratios or indices, absorption and reduced-scattering parameters, water and lipid sensitivity, spatial heterogeneity, and temporal changes in injured tissue.

Develop and validate statistical and machine-learning approaches for burn-depth classification and treatment-decision support, including feature engineering, training and test separation, cross-validation, receiver operating characteristic analysis, sensitivity, specificity, calibration, uncertainty, and subgroup performance.

Correlate optical measurements with surgeon assessments, operative findings, healing outcomes, and pathology-based burn-depth measurements while explicitly evaluating interobserver disagreement and limitations of each reference standard.

Coordinate tissue collection and histology workflows, including biopsy localization, fixation, paraffin embedding, sectioning, H&E, trichrome and picrosirius red staining, whole-slide scanning, collagen-coagulation or necrosis-depth measurement, and quantitative image analysis.

Perform or supervise porcine research procedures relevant to burn creation, imaging, longitudinal monitoring, tissue collection, and protocol compliance in collaboration with veterinary and surgical personnel.

Write and maintain standard operating procedures, equipment logs, calibration records, code repositories, data dictionaries, and analysis documentation sufficient for reproducibility and future regulatory or device-development activities.

Troubleshoot optical, mechanical, electronic, software, synchronization, and data-quality problems; perform root-cause analysis and implement durable corrective actions.

Prepare publication-quality figures, manuscripts, abstracts, presentations, technical reports, grant applications, and progress reports; present work at laboratory meetings and national or international conferences.

Collaborate with burn surgeons, pathologists, optical engineers, physicists, biomedical engineers, statisticians, computer scientists, industry partners, and regulatory or technology-transfer teams.

Train and supervise junior laboratory personnel in optical-system operation, imaging protocols, image analysis, experimental rigor, safety, and data stewardship.

STRONGLY PREFERRED EXPERTISE

Advanced training or experience in biomedical optics, optical engineering, biophotonics, spectroscopy, imaging science, biomedical engineering, physics, electrical engineering, computer vision, or a closely related field.

Hands-on experience designing and operating optical imaging or spectroscopy systems, preferably involving SWIR, NIR, multispectral, hyperspectral, or spatial frequency domain imaging.

Knowledge of light-tissue interactions, including wavelength-dependent absorption, scattering, penetration depth, and contrast from water, lipids, collagen, hemoglobin, and melanin.

Proficiency in MATLAB and Python for image processing, quantitative analysis, data visualization, statistical modeling, automation, and reproducible pipeline development.

Experience with camera control, spectral filters, radiometric or reflectance calibration, image registration, region-of-interest analysis, spectral feature extraction, and optical phantom studies.

Experience with biostatistics and predictive-model evaluation, including nonparametric testing, regression, ROC and AUC analysis, cross-validation, and transparent reporting of model limitations.

Experience in large-animal research, human-subjects imaging, burn or wound research, skin biology, histology, or clinical device studies is highly desirable.

Excellent scientific judgment, troubleshooting, organization, written and oral communication, collaboration, and project-management skills.

PROFESSIONAL EXPECTATIONS

Maintain rigorous, traceable, and reproducible experimental and computational records.

Follow laser and optical safety, electrical safety, biosafety, animal-use, human-subjects, and clinical research requirements.

Maintain a clean, organized, functional workspace and protect sensitive clinical images and data.

Stay current with advances in SWIR detectors, light sources, filters, hyperspectral systems, SFDI, optical-property inversion, fluorescence imaging, computational imaging, and medical-device translation.

Foster a respectful, inclusive, collaborative, and accountable laboratory culture.

BENEFITS

UT Southwestern is proud to offer a competitive and comprehensive benefits package to eligible employees. Our benefits are designed to support your overall wellbeing, and include:

PPO medical plan, available day one at no cost for full-time employee-only coverage

100% coverage for preventive healthcare-no copay

Paid Time Off, available day one

Retirement Programs through the Teacher Retirement System of Texas (TRS)

Paid Parental Leave Benefit

Wellness programs

Tuition Reimbursement

Public Service Loan Forgiveness (PSLF) Qualified Employer

Learn more about these and other UTSW employee benefits!

EXPERIENCE AND EDUCATION

Required

Education

PhD with major course work in the field of assignment or

Master's Degree with major course work in the field of assignment or

Bachelor's Degree with major course work in the field of assignment

Experience

5 years of research experience in the field of assignment with PhD including

2 years at the level of Senior Research Associate or its equivalent or

7 years of research experience in the field of assignment with Master's degree including

3 years at the level of Senior Research Associate or its equivalent or

9 years of research experience in the field of assignment with Bachelor's degree including

4 years Level of Senior Research Associate or its equivalent.

JOB DUTIES

Plans, organizes coordinates, directs, and personally participates in assigned scientific or medical research project. Selects or assists in selection of key research personnel. Determines specific goals or objectives to be attained. Assigns and reviews work of subordinates. Plans experimental protocols.

Reviews and analyzes resulting research data. Revises techniques or approaches to work problems as indicated necessary by research data.

Compiles, writes, and submits research results to supervisor. Prepares or assists in preparation of research papers, reports, and abstracts for publication as one of co-authors.

Designs, develops, or adapts equipment used in experiments or research to obtain desired results.

Plans and supervises training of technicians and other laboratory personnel with respect to proper laboratory techniques, use of laboratory equipment, and safety procedures.

Supervises group of employees of lower grade engaged in performance of technical and professional work, which vary from routine to complex laboratory work tasks.

May confer with industrial, governmental, or other groups concerning progress or results of research project.

Performs other duties as assigned.

SECURITY AND EEO STATEMENT

Security

This position is security-sensitive and subject to Texas Education Code 51.215, which authorizes UT Southwestern to obtain criminal history record information.

EEO

UT Southwestern Medical Center is committed to an educational and working environment that provides equal opportunity to all members of the University community. As an equal opportunity employer, UT Southwestern prohibits unlawful discrimination, including discrimination on the basis of race, color, religion, national origin, sex, sexual orientation, gender identity, gender expression, age, disability, genetic information, citizenship status, or veteran status.

Notes for ECS 1100 students

How this connects to your major

Biomedical Engineering: device design, clinical use, testing, quality, human factors or biomedical research. These are teaching annotations, not an employer guarantee of eligibility.

A path to explore after further study; graduate-level requirements are retained.

BME: Named discipline appears in source; consult qualifications for eligibility

Skills mentioned in the posting

Python, MATLAB, Machine learning, Statistics, Data visualization, Verification/validation, Communication/teamwork

Eligibility excerpts not shown above

Compensation (source excerpts): Not stated in captured posting text.

Course sheet tags

Majors (teaching tags): BME; Career stage (teaching tag): Research / graduate training path; Geography: DFW; Recruiting status at capture: Posting readable at capture; no closure notice found; Employment type: Full-time.

Source record

Original posting https://jobs.utsouthwestern.edu/job/23767000/research-scientist-biomedical-optics-levi-lab-dallas-tx/
Accessed UTC: 2026-10-07T20:11:31.745316+00:00
Saved raw source: sources/postings/utsw_23767000.json
SHA-256: 4b7e7265a39a8d92bf362cbe81c7455f8bb08e50f1f631945ff9f91b1d3a7035